Experiment - 2.

AIM:- Convert the ER diagram into a relational schema, write Complete My SQL, CREATE TABLE Statement with primary keys, foreign key, NOT NULL, UNIQUE, ON DELETE CASCADE and SET NULL Constraints. Insert sample data and demonstrate referential integrity violations.

RELATIONAL SCHEMA. →

Customer(
CustomerID, PK,
Name,
Email UNIQUE,
Phone No,
DOB.
)

Seller(
Seller ID PK,
Seller Name.
Phone No UNIQUE
)

Address(
Address ID, PK,
Customer ID FK,
PINCode,
State,
City
)

Category(
Category ID PK,
Category Name UNIQUE,
Description
)

Product (
Product ID PK,
Category ID FK,
Price,
Image.
)

Fashion (
Product ID PK
FK
)

Order (
Order ID PK,
Customer ID FK,
Order Date,
Order Status,
)

Payment
Payment ID PK.
Order ID FK UNIQUE,
Payment Date,
Amount,
Payment method,
Status,
)

Electronic (
Product ID PK,
FK.
)

Books (
Product ID PK,
FK,
Seller ID FK.
)

Order_item (
Order ID PK | FK
item_seq_NO PK,
Product ID FK,
Quantity,
Unit_Price
)

Delivery (.
Delivery ID PK,
Order ID FK UNIQUE
Courier Name.
Status.
Delivery Date.
)

SOURCE CODE :-

```
CREATE DATABASE Ecommerce DB;
  USE Ecommerce DB;

CREATETABLE Customer (
            Customer ID INT Primary Key Auto_Increament,
            Name Varchar(100) NOT NULL,
            Email Varchar (100) NOT NULL UNIQUE,
            Phone No Varchar (15) NOTNULL,
            DOB DATE NOT NULL.

  );
CREATE TABLE Address (
        Address ID INT primary key Auto_ increament.
        Customer ID INT NOTNULL,
        PINCode Varchar (10) NOTNULL,
        State Varchar (50) NOTNULL.
        City Varchar (50) NOTNULL,
       FORIEGN KEY (customer ID) references Customer.
       (Customer ID) on DELETE CASCADE.
  );
    CREATE TABLE Category,
            Category ID, INT Primary auto-increament,
        Category Name Varchar(100) NOTNULL UNIQUE,
        Description Varchar (255),

    );

    CREATE TABLE Seller.(
        Seller ID INT Primary key auto_increament,
        Seller Name Varchar(100) NOTNULL,
        Phone No Varchar (15) NOTNULL UNIQUE
    );
```

```
CREATE TABLE Product (
    ProductID INT Primary Key auto_increament,
    CategoryID INT NOTNULL
    Price Decimal (10,2) NOTNULL,
    Image varchar (255),
    Foreign key (CategoryID) refrences Category
    (CategoryID) ON DELETE CASCADE )

);

CREATE TABLE Orders(
    OrderID INT Primary auto_increamat.
    Customer INT NOTNULL,
    OrderDate DATE NOTNULL,
    OrderStatus Varchar (50) NOTNULL,
    Foreign Key (CustomerID) refrences Customer
    (CustomerID) on DELETE CASCADE)

);
CREATE TABLE OrderItem(
        OrderID INT NOTNULL,
        item_seq_NO INT NOTNULL,
        ProductID INT NOTNULL,
        Quantity INT NOTNULL,
        Unit_Price DECIMAL (10,2) NOTNULL,
        Primary Key (OrderID, item_seq-NO),
    Foreign key (OrderID) refrences orders (OrderID)
            ON DELETE CASCADE
    Foreign key (ProductID) refrences product (productID)
            ON DELETE CASCADE

);
```

```
CREATE TABLE Payment (
    Payment ID INT Primary key auto-increament
    Order ID INT NOT NULL UNIQUE,
    Payment Date Date NOT NULL,
    Amount Decimal (10,2) NOT NULL,
    Payment method Varchar (30) NOT NULL,
    Status Varchar (30) NOT NULL,
    Foreign Key (order ID) references order
                                    (order ID)
        ON DELETE CASCADE
);

CREATE TABLE Delivery (
    Delivery ID INT Primary key auto-increament
    order ID INT NOT NULL UNIQUE,
    CourierName Varchar (100) NOT NULL,
    Status Varchar (30) NOT NULL,
    Delivery Date DATE,
    Foreign Key (order ID) references order
                                    (order ID)
        ON DELETE CASCADE
);
```

Inserting Sample Data.

Insert into Customer (Name, Email, Phone No, DOB) VALUES
("[redacted] Naman", Naman@gmail.com", "7374217295", 2005-05-15"),
("Arjun", "Arjun@gmail.com", "9336677999", "2004-08-20");

Insert into Category (category Name, Description) VALUES.
("Electronics", "Electronic product");
("Fashion", "Fashion product"), ("Books", Books"

Insert into Seller (Seller Name, Phone No) VALUES.
("Tech store", 9000000001"),
("Book store", 9000000002");

Insert into product (categoryID, Price, Image) VALUES
(1, 50000, Laptop.jpg"), (2, 999, "Shirt.png"),
(3, 299, book.jpg");

Insert into Electronic values (1);
Insert into Fashion values (2);
Insert into Books values (3, 2);
Insert into address (Customer ID, Pincode, State, city)
VALUES.
(1, "208001", Uttar pradesh", Kanpur"),
(2, "110001", "Delhi", "Delhi");

Insert into orders (customerID, order Date, Order
-Status) VALUES.
(1, "2026-08-19", "Placed"),
(2, "2026-08-19", "Shipped");

Insert into OrderItem VALUES.
(1,1,1,1,1,50000), (2,1,2,2,999);

Insert into Payment (orderID, PaymentDate, amount,
Payment method, status) VALUES.
(1, "2026-08-19", 50000, "UPI" "Success"),
(2, "2026-08-19", 1998, "Card", "Success");

Insert into Delivery (orderID, CourierName, status,
DeliveryDate)
VALUES.
(1, "Delivery", "Delivered", "2026-08-20"),
(2, "DTDC", Pending", NULL);

→ Referential Integrity violations :-
Insert into Orders (customerID, orderDate,
orderstatus).
VALUES.
(999, "2026-08-19", "Placed");

OUTPUT: →

ERROR - Cannot add or update a child row : a foreign key constraint fails.

Unique Constraints violation :-

Insert into customer (Name, Email, PhoneNo, DOB)
VALUES.
("Test", "Naman Sahil@gmail.com", "9999999999",
"2025-01-01");

OUTPUT →

Error :- Duplicate entry 'Sahil@gmail.com' for key 'to Customer.Email'.

Select * from Customer;

Select * from Product;

Select * from Orders;

Select * from Order Item;

ON DELETE SET NULL.

DELETE from Seller where Seller ID = 2;

Select

Select * from ~~Books~~ Order where Customer ID = 2;

OUTPUT →

Customer ID	Name	Email	Phone No	DOB.
1	Naman	Naman@gmail.com	9897401240	2005-05-15
2	Arjun	Arjun@gmail.com	9336679089	2004-08-20

Product ID	Category ID	Price	Image.
1	1	50000.00	laptop.png
2	2	999.00	shirt.png
3	3	599.00	book.png.

order ID	customer ID	order Date	order status
1	1	2026-08-19	Placed
2	2	2026-08-19	shipped

order ID	item_seq_NO	productID	quantity	unit_price
1	1	1	1	50000.00
2	1	2	2	999.00

Product ID	seller ID.
3	NULL.

Empty set